

The Generalized-Sturmian Secular Equation Block-Encodes Without a Metric: A Quantum-Algorithm Analysis, Metric-Free for Atoms and a Conditioning Frontier for Molecules*

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Central result. In Avery’s isoenergetic (potential-weighted) posing, the *atomic* Coulomb–Sturmian secular equation is a standard, metric-free eigenproblem—an apparently novel quantum-encoding target, with both documented cost-risks of the genre (metric conditioning; an outer energy search) structurally absent—whereas the naive L^2 posing of the same basis inflates the cost instead. Its block-encoding 1-norm grows *more slowly than the configuration count* over every basis we can compute, and *more* slowly as the basis grows. That advantage, however, belongs to the *basis-growth rule* rather than to the construction: it holds when $l_{\max}$ is fixed and higher- n configurations are added, and reverses to superlinear when the basis is grown as full hydrogenic shells. The cheap rule also carries an accuracy floor, whose mechanism we locate exactly: the metric-free form holds *iff* the basis scale is locked to the eigenvalue, $\lambda = p_\kappa = \sqrt{-2E}$, and that lock is not the variational optimum. Solving the *same* span variationally with the scale freed reaches chemical accuracy, but hands back *both* of the cost risks named above — the L^2 metric returns, and the scale becomes an outer search — along with the encoding advantage itself (1-norm growth from $K^{0.72}$ to $K^{2.75}$ on the s -only comparison ladder, $\text{cond}(S) \sim K^{0.94}$). The absence of those two risks and the presence of the floor are therefore not independent virtues and defects but one structural fact. The floor is moreover a *ground-state* pathology rather than a property of the method: because $\|M\|_1$ does not depend on which root is extracted, every 1S state costs the same to encode, and at the largest computed basis $K = 452$ the ground state sits $4.28\times$ above chemical accuracy while 2^1S sits $1.08\times$ above it, the posing cost falling by $4.3\times$, $3.0\times$ and $2.6\times$ across the first three rungs of the 1S ladder. The result is therefore a measured trade-off between cost growth, attainable accuracy and *which state is targeted*, not an unconditional efficiency claim. The development follows.

A recurring intuition in quantum simulation is that a Coulomb–Sturmian / hyperspherical basis is well suited to a quantum computer, because its angular matrix elements are exact rationals generated on the device from quantum-number labels rather than loaded as a classically-computed tensor; only the one-dimensional radial content need be shipped in. We test the load-bearing obstruction to that idea—the cost of *encoding* the basis—and find it turns sharply on how the basis is posed. **[MEASURED]** Encoded naively, in the ordinary L^2 Schrödinger framework, the genuine shared-scale Coulomb–Sturmian basis is L^2 -non-orthogonal; a second-quantized (Jordan–Wigner) encoding must Löwdin-orthogonalize it, and the LCU 1-norm λ then *inflates* as $\lambda \sim Q^{3.3}$ (versus $Q^{1.2}$ for an orthonormal hydrogenic basis), driven by the growing ill-conditioning of the overlap. **[ESTABLISHED, from Avery’s formulation]** The generalized-Sturmian *isoenergetic* reformulation removes this obstruction for atoms: posed in the potential-weighted metric in which the Coulomb–Sturmians are orthonormal, the atomic secular equation is a *standard* (metric-free) eigenproblem $[\text{diag}(ZR_\nu) + T' - p_\kappa \mathbb{I}]B = 0$ whose eigenvalues are the scaling parameters $p_\kappa = \sqrt{-2E}$ directly and whose interelectron matrix T' is a matrix of pure numbers, independent of p_κ , of energy, and of Z . **[MEASURED]** We implement it, validate it against helium (a single Goscinskian configuration reproduces the textbook variational value -2.847 Ha; adding configurations lowers the energy monotonically through the s , $+p$ and $spdf$ sectors—past the s -sector limit -2.873 Ha to -2.897 Ha, toward the exact non-relativistic -2.90372 Ha), and measure the decisive quantity: the block-encoding 1-norm of the secular matrix grows *more slowly than the matrix dimension*, $\|M\|_1 \sim K^{0.82}$ over the computed range $K = 74\text{--}164$ (full $s+p+d+f$, helium; $K^{0.73}$ s -only)—the opposite of the naive inflation. The exponent is a fit over a window and not an asymptotic regime: its local slope *falls* monotonically with K (Sec. IV), it moves with the nuclear charge, and it belongs to the fixed- $l_{\max}$ growth rule rather than to the construction. **[OBSERVATION]** A targeted prior-art search finds no existing quantum algorithm combining a Sturmian basis, a quantum eigenvalue routine, and the isoenergetic inversion; the two documented cost risks for such an algorithm—metric conditioning and an outer energy search—are both *absent* in the atomic case (no metric; the eigenvalues are the energies). **[MEASURED]** For molecules the Shibuya–Wulfman metric returns and the problem is again generalized; we measure its conditioning and find it uniformly better than the L^2 overlap—its intra-center block is exactly the identity—and markedly better at larger internuclear separation, but still growing with basis size through the inter-center coupling. **[SYMBOLIC + MEASURED]** The growth law is derived (a band-limited concentration rate, $1 - \sigma_{\max} \propto (kR)^2/n^2$, giving asymptotic exponent 2—the leading order is derived, the remainder measured rather than bounded; the fitted window exponents 1.85/1.97 are pre-asymptotic readings). **[MEASURED]** The same cross-center singular spectrum carries the composition-wall commutator, whose norm

saturates at $\frac{1}{2}$. **[RESOURCE MODEL]** Turning this into a block-encoding count for H_2^+ (validated to bind, 1.3% s -only), the payoff is the gerade/ungerade lever: the σ_g ground state lives in the gerade sector, whose conditioning is *flat*—and flat at an exact constant of the sinc symbol, $2/(1 + \min_x j_0(x)) = 2.555041\dots$, independent of R and of basis size—so the ground-state metric penalty does *not* grow with basis size—a $\sim 60\times$ reduction versus the full metric and $10^3\text{--}10^4\times$ smaller than a standard Gaussian metric for the same problem (a metric-versus-metric ratio that scales with the Gaussian exponent spread, hence illustrative rather than a fixed factor), at ~ 30 qubits. **[MEASURED]** The penalty also does not compound with electron number: the metric is discharged once at the one-electron molecular-orbital construction, after which the many-electron secular equation carries an identity metric—the metric cost is one-electron, not N -electron. **[OPEN]** The atomic case is a clean, apparently novel, metric-free quantum secular equation; the molecular case is a genuine frontier in which the metric cost is mitigated for excited/ungerade states, essentially removed for the ground state of *equivalent-center* systems (a water probe shows a symmetry-inequivalent heavy center reinstates it), and confined to one electron. **[MEASURED]** Its remaining question—whether the N -electron 1-norm stays sublinear as it does for atoms—we answer in the negative: the atomic sublinearity rides on a diagonal nuclear matrix $T^0 = ZR_\nu$ that is single-center-specific, and a validated two-center CI (dissociating H_2 correctly) has a *standard* block-encoding 1-norm growing polynomially ($\sim n_{\text{orb}}^{2.2}$), with no sublinear behavior. The real molecular savings are the metric levers (gerade sector, large separation, one-electron-only metric), not a sublinear matrix. We do not demonstrate the algorithm on hardware; this is a structural and resource analysis.

I. INTRODUCTION

The generalized-Sturmian method of Avery and Avery [1–4], built on Goscinski’s weighted wave equation and the momentum-space Fock projection, is one of the most elegant classical frameworks for atomic and molecular electronic structure. Its angular content is closed-form—Gaunt integrals, Wigner $3j/6j$ symbols, Shibuya–Wulfman integrals—generated from the quantum numbers (n, ℓ, m) with no spatial quadrature. This is the same discrete angular skeleton that the Geometric Vacuum program reads as a graph (Papers [15, 16]); the two frameworks meet exactly at those closed-form angular integrals.

The angular cleanness suggests a quantum-computing use, articulated independently in the Sturmian and Geometric-Vacuum settings: the bottleneck in quantum simulation is often not the term count of the Hamiltonian but the *input/output*—classically computing a two-electron integral tensor and loading it onto the device. If the angular structure can be generated on the device from labels, and only the one-dimensional radial factors loaded, the loading cost is dominated by a small radial-seed count rather than an $O(M^4)$ tensor. This is a known-good lever in fault-tolerant simulation: plane-wave quantum-simulation methods compute their coefficients rather than loading them and win decisively for large basis sets [7, 8] (first quantization, with $O(\eta \log N)$ qubits, at the [8] end).

The lever has a well-documented Achilles heel, however. Its payoff is governed not by the loading count alone but by the *LCU 1-norm* $\lambda = \sum_i |c_i|$ of the encoded Hamiltonian, which sets the query and T -gate cost of qubitization and block-encoding [9, 10]; plane

waves win the I/O yet pay a large λ . The question this paper addresses is whether a Coulomb–Sturmian encoding keeps λ bounded. We find that the answer depends entirely on *how the basis is posed*, and that the isoenergetic reformulation—the defining move of the generalized-Sturmian method—is precisely what makes the atomic problem cheap.

The paper is organized as the argument runs. Section II measures the naive obstruction. Section III states the isoenergetic reformulation and why it is metric-free for atoms. Section IV implements and validates it on helium and measures the sublinear 1-norm. Section V states the quantum algorithm and its novelty. Section VI measures the molecular Shibuya–Wulfman conditioning. Section VII states the honest boundary.

II. THE NAIVE OBSTRUCTION: 1-NORM INFLATION FROM L^2 ORTHOGONALIZATION

A second-quantized (Jordan–Wigner) encoding requires an L^2 -orthonormal set of single-particle spin-orbitals. The genuine shared-scale Coulomb–Sturmians are *not* L^2 -orthonormal: at a common scale k their higher members grow linearly dependent, so a qubit encoding must first orthogonalize them (Löwdin, $S^{-1/2}$), and the coefficient distribution—hence $\lambda = \sum_i |c_i|$ —spreads.

[MEASURED] In a controlled s -sector helium construction in which only the radial convention is changed and the ground shell is anchored so the two families coincide at $Q = 2$ qubits, the LCU 1-norm scales as

$$\lambda_{\text{hydrogenic}} \sim Q^{1.19}, \quad \lambda_{\text{Sturmian}} \sim Q^{3.33}, \quad (1)$$

a factor of ~ 18 larger by ten qubits (a measured ratio from the $\lambda(Q)$ data; like Eq. (6) these exponents are numerical fits, and the tracked claim is the sublinear-vs-superlinear contrast, not the precise value). The

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mechanism is the L^2 overlap S : with helium Coulomb–Sturmians its condition number climbs (3.0, 5.8, 13.9, 32.2 for $N = 2, 3, 5, 8$), and the dense $S^{-1/2}$ built from an ill-conditioned Gram inflates λ . This is a currency-change of the familiar cost-conservation phenomenon in non-orthogonal quantum chemistry (the overlap metric never disappears; moving it densifies operators, states, or coefficients [15]). Encoded this way, the Sturmian basis is the *worst*, not the best, choice.

[MEASURED] The obstruction is intrinsic to the L^2 posing rather than to a choice of orthogonalization: the L^2 Löwdin encoding inflates the LCU 1-norm as $\lambda \sim Q^{3.33}$ against $Q^{1.19}$ for an orthonormal hydrogenic basis (Eq. (1)), measured through a faithful Jordan–Wigner LCU. That is the cost obstruction, and it stands. *Correction, 2026-09-07 — a stronger claim is withdrawn here.* Earlier versions added that re-solving the converged $s+p+d+f$ helium problem as the generalized eigenproblem $MB = pSB$ *diverges*, with the metric condition number climbing $4 \rightarrow 3673$ and the energy running to -3×10^6 Ha. Both numbers are artifacts of the fixed 60-bohr radial domain (Appendix A). On a converged domain $\text{cond}(S) = 16.0$ at the same point, and the growth is mild and smooth, $\text{cond}(S) \approx 0.12 K - 4.07$ at $K = 10$, 23.5 at $K = 164$, 56.3 at $K = 452$. The blow-up switches on exactly where n_{max}^2 first exceeds the domain size, and every domain agrees to four digits below that point. The L^2 Gram matrix of this basis is therefore ordinary, not ill-conditioned, and we no longer claim the metric-free posing is the *only* well-conditioned one. What we do claim is the cost statement above: the L^2 posing is expensive to *encode*, which is a different assertion from being numerically unstable, and is the one this paper’s argument needs.

III. THE ISOENERGETIC REFORMULATION IS METRIC-FREE FOR ATOMS

The Coulomb–Sturmians obey a *potential-weighted* orthonormality relation [3, 5, 6],

$$\int d^3x \varphi_{n'\ell'm'}^*(x) \frac{1}{r} \varphi_{n\ell m}(x) = \frac{k}{n} \delta_{n'n} \delta_{\ell'\ell} \delta_{m'm}, \quad (2)$$

orthonormal in the $1/r$ weight, *not* in L^2 . **[MEASURED]** We confirm Eq. (2) numerically: the potential-weighted overlap is diagonal to $\sim 10^{-6}$ off-diagonal, whereas the L^2 overlap of the same basis is the ill-conditioned matrix of Sec. II.

Building the electronic-structure problem in this native metric is the generalized-Sturmian method. With the basis potential taken to be the bare nuclear attraction $V_0 = -\sum_j Z/r_j$, the Goscinskian configurations Φ_ν are Slater determinants of hydrogenlike orbitals sharing one weighted charge $Q_\nu = p_\kappa/R_\nu$, with configuration root $R_\nu = \sqrt{\sum_j 1/n_j^2}$, all isoenergetic at $E = -\frac{1}{2}p_\kappa^2$. The kinetic term cancels against V_0 by construction, the

nuclear-attraction matrix is diagonal, $T_{\nu'\nu}^0 = ZR_\nu \delta_{\nu'\nu}$, and the atomic secular equation is [3, 4]

$$\sum_\nu \left[ZR_\nu \delta_{\nu'\nu} + T'_{\nu'\nu} - p_\kappa \delta_{\nu'\nu} \right] B_\nu = 0, \quad E = -\frac{1}{2}p_\kappa^2. \quad (3)$$

Three features of Eq. (3) are load-bearing for a quantum algorithm, and all are consequences of the potential-weighting:

1. It is a *standard* eigenproblem, $MB = p_\kappa B$ with $M = \text{diag}(ZR_\nu) + T'$. There is *no overlap metric* S : the potential-weighting has absorbed it into the diagonal nuclear term. The ill-conditioning of Sec. II is not repaired, it is *never present*.
2. Its eigenvalues are the scaling parameters p_κ *directly*; one diagonalization yields the whole spectrum via $E = -\frac{1}{2}p_\kappa^2$. There is no self-consistency and no outer search over the energy.
3. The interelectron matrix $T'_{\nu'\nu} = -p_\kappa^{-1} \int \Phi_{\nu'}^* V' \Phi_\nu$ is a matrix of *pure numbers*, independent of p_κ , of E , and of Z : the two-electron integrals scale as $Q_\nu = p_\kappa/R_\nu$, and the $-p_\kappa^{-1}$ prefactor cancels the p_κ . A single T' serves an entire isoelectronic series.

T' is assembled by Slater–Condon reduction of the two-electron operator, with the $1/r_{12}$ Laplace expansion giving a finite multipole sum (the angular factors are products of Gaunt integrals / Wigner $3j$, recoupling to $6j$ for coupled configurations) times a radial Slater integral [4].

IV. ATOMIC IMPLEMENTATION, VALIDATION, AND THE SUBLINEAR 1-NORM

We implement Eq. (3) for helium ($Z = 2$). For the leading configuration $1s^2$ the matrix is 1×1 : $R_{1s^2} = \sqrt{2}$, the unit-charge integral is $\langle 1s^2 | r_{12}^{-1} | 1s^2 \rangle = \frac{5}{8}$, so

$$T'_{1s^2, 1s^2} = -\frac{5}{8}/\sqrt{2} \approx -0.442, \quad p_\kappa = Z\sqrt{2} + T' = 2.386, \quad (4)$$

giving $E = -p_\kappa^2/2 = -2.847$ Ha. **[MEASURED]** Computed on a radial grid this reproduces the textbook variational helium value (-2.84766 Ha; the pure number $\frac{5}{8}/\sqrt{2}$ is confirmed to grid precision), and the “bare” matrix (no V') returns -4.0 Ha, i.e. two He^+ $1s$ orbitals—the exact non-interacting limit. **[MEASURED]** Adding configurations coupled to total 1S lowers the energy monotonically,

$$-2.847(1s^2) \rightarrow -2.873(s) \rightarrow -2.894(+p) \rightarrow -2.897(spdf, K=$$

converging toward the exact non-relativistic -2.90372 Ha. **[MEASURED]** The residual is *not* a deficiency of the configuration span. A variational CI over the *identical* span, with the global scale optimized, reaches 1.28 mHa of the exact energy at $K = 130$, where the isoenergetic solution of that same span

stalls at 7.46 mHa; in the s -sector, where the exact s -limit -2.879029 Ha is known independently, the same comparison is 0.15 mHa against 4.43 mHa at $K = 136$. The residual is therefore the price of the *posing*, not of the basis: the metric-free form requires the basis scale to be locked to the eigenvalue, $\lambda = p_\kappa$, and that lock is not the variational optimum. **[INTERNAL THEOREM]** The same identity makes the bound automatic rather than fortunate—the isoenergetic root is the lowest root of the fixed-scale generalized problem $H(p_\kappa)C = ESC$, so every point lies above the exact value by construction, not by observation. **[OBSERVATION]** The canon does not appear to state this. Its variational discussion is framed on the Rayleigh–Ritz principle for the *scanned* k -dependent determinant and does not partition the argument between the locked and scanned postings; a targeted consultation returned no statement that the fixed-scale metric-free problem bounds from above. We therefore claim it.

[ESTABLISHED, from Avery] The mechanism is Avery’s, not ours. The helium ground state needs *in-out* radial correlation, which variational treatments supply through split-shell ($1s\,1s'$) functions carrying two independent exponents; a Goscinskian $1s^2$ configuration places both electrons at $n = 1$ and therefore at the identical exponent Q_ν/n_j —for *any* weighting potential, since a separable V_0 still assigns one β_ν per configuration. Excited configurations obtain two scales free from $n_a \neq n_b$, and the state-dependence this predicts is measurable: the posing cost is 4.21 mHa for the ground state against 0.98 mHa for 2^1S at $K = 105$, and the ratio in fact *narrows* slowly with basis size — 5.17 at $K = 36$, 4.66 at 78, 4.29 at 105 — while the absolute separation widens ($2.84 \rightarrow 3.23$ mHa) and then flattens. The floor is a ground-state pathology rather than a property of the method.

[INTERNAL THEOREM] *Selection cannot rescue the locked posing.* Because T' is a matrix of pure numbers, its entries do not depend on which *other* configurations are present, so the secular matrix of any subfamily A is *exactly* the corresponding principal submatrix of M (verified bit-identically). Cauchy interlacing then gives $\lambda_{\max}(M_A) \leq \lambda_{\max}(M)$, and since $E = -p_\kappa^2/2$ with $p_\kappa = \lambda_{\max} > 0$,

$$E(A) \geq E(M) \quad \text{for every sub-family } A \subseteq M, \quad (5)$$

the families being nested in $n_{\max}$ so that E decreases monotonically along the ladder. No choice of configurations, however optimized, beats the family it is drawn from. **[MEASURED]** The best 102 by ground-state weight out of a $K = 244$ pool returns 7.25 mHa against that pool’s own 7.06 mHa — *above* it, as Eq. (5) requires — while of 200 random 102-subsets the best reaches only 13.9 mHa and the worst 903 mHa — 58% of them omit the dominant $1s^2$ configuration entirely. Note what supplies the bound: it is the same pure-number property that makes the encoding attractive. Were T' basis-dependent, sub-families would not be principal submatrices and in-

terlacing would not apply.

[OBSERVATION] A comparison to the -2.90250 Ha reached with 102 optimized Coulomb–Sturmian configurations [2] is therefore *withdrawn*, and on stronger grounds than a mismatch of numbers: by Eq. (5) that figure (1.2 mHa short) *cannot* have come from the locked-scale Goscinskian posing, at any K or under any selection — our largest computed family stands at 6.82 mHa and the ladder is monotone. It matches instead our scale-optimized ladder (1.64 mHa at $K = 100$). **[OBSERVATION]** Consultations of the primary canon, relayed, settle the attribution — and one of them supplies an external test of Eq. (10). The posing was identified first: the scaling parameter *scanned as a free variational parameter and minimized*, unlinked from the output eigenvalue, with the ground state’s difficulty attributed there, as here, to $1s^2$ forcing both electrons to $Q_\nu = p_\kappa/\sqrt{2}$. The same account also described that calculation as evaluating a *metric-free* matrix at each step — which Eq. (10) forbids, since projecting a basis built at E_{basis} against a target E_{out} leaves the L^2 overlap with coefficient $(E_{\text{basis}} - E_{\text{out}})$, and unlinking the scale is precisely what makes that coefficient nonzero. Put back to the source together with the projection, the derivation was accepted and the description corrected: the calculation was an *ordinary variational* configuration interaction, $HC = ESC$, and the metric-free cancellation belongs exclusively to the locked isoenergetic method. We record the attribution as secondary-source — it is not a quotation, and a primary-source check remains the one thing that could overturn this section — but the episode is worth stating for what it is: an external description of a calculation as simultaneously unlinked and metric-free, posed against Eq. (10) and withdrawn. The machinery is correct, including the higher- ℓ angular correlation (Gaunt / Wigner-3j multipole coupling).

The decisive quantity is the block-encoding cost. **[MEASURED]** The entrywise 1-norm of the secular matrix, $\|M\|_1 = \sum_{\nu'\nu} |M_{\nu'\nu}|$ (an upper bound on the achievable LCU λ), grows *more slowly than the matrix dimension* K ,

$$\|M\|_1 \sim K^{0.82} \quad (\text{full } s+p+d+f, \text{ helium, } K = 74\text{--}164), \quad (6)$$

the opposite of the naive inflation Eq. (1).

Three qualifiers travel with that number, and none is cosmetic.

[MEASURED] *First, it is not a stable exponent.* The fit is over a window, and the window’s value is not the object’s behaviour. Continuing the same family, the *local* slope falls monotonically: 0.827, 0.818, 0.810, 0.802, 0.794, 0.787, 0.781, 0.776, 0.771, 0.766 across $K = 100\text{--}514$, with no plateau at the largest basis we can reach. A two-parameter $cK^p \log K$ form fits the data an order of magnitude better than any pure power law, for every norm family we measured. The honest statement is that $\|M\|_1$ grows sublinearly over every computable basis and moves *away* from linear as the basis grows; any single exponent is a property of the window it was fitted

in.

[MEASURED] *Second, it is a helium number.* T' is Z -independent by construction, so $\|M\|_1$'s exponent is a mixture whose weight moves with the nuclear charge: the same window gives 0.947 at $Z = 1/2$, 0.819 at $Z = 2$, and 0.717 at $Z = 30$. Every exponent in this paper is quoted for helium.

[MEASURED] *Third — and this is the substantive finding — the sublinearity is a property of how the basis is grown, not of the isoenergetic construction.* The family used here holds $l_{\max}$ fixed and adds higher- n configurations, whose couplings decay as n^{-2} . Grown the other way, as full hydrogenic shells with $l_{\max} = n_{\max} - 1$, the same secular construction gives $\|T'\|_1 \sim K^{1.07}$ and a total exponent that rises with K ($0.867 \rightarrow 0.911$ over $K = 56$ – 220). Superlinear behaviour and a rising exponent are therefore real — for a different growth rule. What Eq. (6) measures is the fixed- $l_{\max}$ rule, and the cost advantage belongs to that rule rather than to the method.

[MEASURED] *Which block carries it.* Splitting $M = \text{diag}(ZR_\nu) + T'$ and fitting each over $K = 74$ – 340 on a converged radial domain:

$$\|T^0\|_1 = Z \sum_{\nu} R_{\nu}, \quad \|T'\|_1 \sim K^{0.88}, \quad \|T'\|_1^{\text{off}} \sim K^{0.94}. \quad (7)$$

The three are related by the exact identity $\|M\|_1 = \|T^0\|_1 - \|\text{diag } T'\|_1 + \|T'\|_1^{\text{off}}$. **Neither T' grouping is superlinear:** the full block's local slope crosses 1 near $K \approx 37$ and falls to 0.805 by $K = 514$; the off-diagonal part crosses near $K \approx 91$ and falls to 0.841. The nuclear diagonal is nonetheless the slowest-growing block, and its share of the norm falls monotonically (64.9% at $K = 74$ to 57.0% at $K = 340$), so Eq. (6)'s exponent sits between the two and drifts with their mixture. This is the same structural fact Sec. VI reaches from the other side: the advantage rides on the clean single-center diagonal T^0 , which is exactly what two centers destroy. Angular structure dilutes but does not break the contrast — the s -only exponent 0.73 rises to 0.82 with p, d, f present, on matched windows.

[SYMBOLIC] *The nuclear diagonal is not a power law at all.* $\|T^0\|_1 = Z \sum_{\nu} R_{\nu}$ requires no two-electron integral and no radial grid: it is a closed-form sum over the configuration list, with $K = 2N^2 - 4N + 4$ exactly for $l_{\max} = 3$. Its asymptotics are

$$\|T^0\|_1 \sim Z \sqrt{2K} \ln(K/2), \quad (8)$$

so its log-log slope is $\frac{1}{2} + O(1/\log K)$ — asymptotic exponent $1/2$, not a fitted power. Evaluated on the exact sum the slope runs 0.709 at $K = 20$, 0.705 at 164, 0.662 at 12484 and 0.603 at 498004, still falling. A fit inside $K = 74$ – 164 returns 0.705; that is what the window yields, not what the object does. Eq. (8) is the one leg of this section that is derived rather than measured, and it is grid-independent by construction.

[MEASURED] *What the sublinearity costs.* The fixed- $l_{\max}$ family does not converge to the exact energy. On the converged ladder its ground-state error is 8.04 mHa at $K = 74$ and 6.78 mHa at $K = 514$; fitting $\Delta E(K) = \Delta E_{\infty} + bK^{-q}$ gives $\Delta E_{\infty} = 6.44$ mHa (residual < 0.002 mHa), with 6.446 mHa still predicted at $K = 10^5$. Seven times the basis closes 16% of the deficit. The floor is $4.0\times$ chemical accuracy (1.594 mHa).

[INTERNAL THEOREM] *Where the floor comes from.* It is neither angular truncation nor a deficiency of the span. Write the configurations at a free global scale λ , which the Goscinskian construction fixes to p_{κ} . Because each Φ_{ν} is hydrogenic at $Q_{\nu} = \lambda/R_{\nu}$, the one-body Coulomb metric is exactly diagonal,

$$W_{\mu\nu} \equiv \left\langle \Phi_{\mu} \left| \sum_j r_j^{-1} \right| \Phi_{\nu} \right\rangle = R_{\nu} \delta_{\mu\nu}, \quad (9)$$

verified entrywise to 5×10^{-11} . Hence the kinetic matrix at unit scale is $T = \mathbb{K} - S/2$, and the variational problem over the same span is $H(\lambda)C = E SC$ with $H(\lambda) = \lambda^2(\mathbb{K} - S/2) + \lambda(-Z \text{diag } R_{\nu} + G)$ and $G_{\mu\nu} = \langle \Phi_{\mu} | r_{12}^{-1} | \Phi_{\nu} \rangle$. Putting $E = -\lambda^2/2$ cancels every S term *identically* and leaves $(Z \text{diag } R_{\nu} - G)C = \lambda C$, which is Eq. (3). The metric-free posing is therefore the variational problem of the same span, evaluated at the one scale where the L^2 metric drops out:

$$\text{metric-free} \iff E = -\lambda^2/2 \iff \lambda = p_{\kappa} = \sqrt{-2E}. \quad (10)$$

[MEASURED] That lock is not the variational optimum. Freeing λ over the identical span reaches 0.15 mHa of the exact s -limit at $K = 136$ (s -only, where -2.879029 Ha is known independently) against 4.43 mHa locked, and 1.28 mHa of the exact energy at $K = 130$ ($spdf$) against 7.46 mHa locked. The span is not the limitation; the lock is. Two corollaries. The variational bound is automatic rather than fortunate — E_{iso} is the lowest root of $H(p_{\kappa})C = E SC$, so no point can fall below the exact value. And freeing the scale forfeits the encoding advantage entirely. On the s -only comparison ladder ($K = 21$ – 136 ; note this is a different sector and window from the headline exponent of Eq. (6), so the four numbers below are quoted as one matched set) the locked 1-norm's $K^{0.72}$ becomes $K^{1.95}$ for $H(\lambda^*)$ and $K^{2.75}$ for the whitened $S^{-1/2}HS^{-1/2}$, with $\text{cond}(S) \sim K^{0.94}$ — a factor 5×10^3 at $K = 136$. *Metric-free posing, sublinear 1-norm and accuracy floor are one fact, not three.*

[ESTABLISHED, from Avery] *Why the ground state in particular.* Helium's ground state needs *in-out* radial correlation, supplied variationally by split-shell $1s1s'$ functions carrying two independent exponents. A Goscinskian $1s^2$ configuration places both electrons at $n = 1$ and hence at one exponent Q_{ν}/n_j — for *any* weighting potential, since a separable V_0 still assigns a single β_{ν} per configuration. Excited configurations obtain two scales free from $n_a \neq n_b$. [MEASURED] The state-dependence this predicts is large, and it is free: $\|M\|_1$ is a property of M , not of which root is extracted,

so every $1S$ state costs the same to block-encode and only the delivered accuracy differs. At the largest computed basis, $K = 452$, the ground-state error is 6.82 mHa ($4.28\times$ chemical accuracy) against 1.716 mHa ($1.08\times$) for 2^1S . Reference-free — the posing cost $E_{\text{iso}} - \min_{\lambda} E_{\text{var}}$ requires no known limit — the $1S$ ladder at $K = 105$ reads 4.21, 0.98, 0.32 and 0.13 mHa across the first four roots — reductions of $4.3\times$, $3.0\times$ and $2.6\times$ per rung, themselves shrinking. **[OBSERVATION]** The shrinking Rydberg gaps do not spoil this at chemical accuracy: resolving the k -th root needs $\epsilon_p = \epsilon_E/p_k \approx 8 \times 10^{-4}$ against gaps of 0.336, 0.041, 0.014 and 0.006 — margins of 254, 27, 8.6 and 3.9 — so the query count $\|M\|_1/\epsilon_p \approx 2 \times 10^5$ is state-independent through 4^1S . **[MEASURED]** The remaining cost of an interior root is state preparation, and it is modest. In the L^2 metric the overlap of the dominant single configuration with the true root runs 0.992, 0.798, 0.864 and 0.889 across the first four roots, so 2^1S — not the deeper ones — is the hardest, at $1.25\times$ the ground state in rotations and $1.54\times$ in repetitions; two configurations already reach 0.99. Interior roots become *easier* again, so the driver is mixing at the bottom of the Rydberg series rather than spectral depth. One practical caveat: the dominant configuration does not track the spectroscopic label — the root at He 3^1S is dominated by $(l, n_a, n_b) = (0, 1, 4)$ — so a preparation heuristic keyed to the physical principal quantum number selects the wrong configuration.

[OBSERVATION] *How well the floors themselves are known.* They are bracketed rather than pinned, and the dominant uncertainty is not the one we first looked for. Across 21 fit windows the free-floor value drifts by only 1.3% (ground) and 0.4% (2^1S), and drifts *upward*: the fit approaches from below. A model-free Shanks extrapolation descends from above, so the two together bracket the floor at [6.47, 6.62] and [1.647, 1.676] mHa — the lower endpoint from the fit, the upper from Shanks. Window drift is therefore not the exposure; *model family* is. A two-parameter $c + b/\ln K$ form would place the 2^1S floor *below* chemical accuracy. It is rejected on two grounds rather than one: it fits $96\text{--}342\times$ worse in RMS, and — unlike the three-parameter form, whose limit is stable to three decimals across every window ($1.032, 1.033, 1.033\times$ chemical accuracy) — *its own limit is window-unstable*, running $0.69 \rightarrow 0.79 \rightarrow 0.82\times$ as the window tightens. A model whose asymptote moves with the window is not asserting an asymptote. We nonetheless quote measured ladder endpoints rather than fitted values throughout, and record that “ 2^1S saturates above chemical accuracy” remains a model-selection conclusion rather than a measurement.

[OPEN] Two questions replace the one this section used to raise. Whether a weighting potential of different *radial shape* can reposition the exponent distribution enough to lower the ground-state floor while preserving Eq. (9) — a constant rescaling cannot, since $\beta_\nu Z_w = p_\kappa/R_\nu$ eliminates Z_w identically, so the lever must change the shape of V_0 and not merely its strength. That lever

has a named price, and it is not small: a screened or mean-field V_0 improves the physical quality of each configuration, but losing the $1/r$ form breaks the mapping to hyperspherical harmonics under the Fock projection, and with it the closed-form evaluation of the multi-center and inter-electron integrals — the property this entire encoding rests on. The open question is therefore sharper than “does a better V_0 exist”: it is whether any V_0 of different radial shape *preserves the Fock mapping*. If none does, the trade is a wall with a mechanism rather than unexplored ground. And whether the excited-state operating point survives the state-preparation cost that Eq. (10) says nothing about. As it stands the family is best read as a *fixed-accuracy tier whose accuracy depends on the target state*: sublinear cost at ~ 6.4 mHa for the ground state and ~ 1.8 mHa for 2^1S , bought at the same price.

Scope and comparison. The “sublinear” here is a property of the LCU 1-norm $\|M\|_1$ of a *fixed-size* atomic secular matrix as the configuration count K grows; it is a distinct quantity from the sublinear *gate complexity in the number of plane-wave basis functions* of Babbush *et al.* [24]. Its exponent is in K at *fixed* atomic size — a different axis from the system-size scaling of the electronic-structure 1-norm literature, where the best documented basis-choice lever, localized orbitals [23], reaches $\sim N^{1.34}$ in the orbital count and is still superlinear; the two are a regime contrast, not a like-for-like exponent comparison. Two further caveats: the construction uses L^2 -normalized configurations rather than the exact potential-weighted normalization, and K is the matrix dimension, not the qubit count. Given that the exponent has already moved once under a change of numerical domain, we no longer assert that the normalization would leave it fixed; that is an open check.

Provenance of the numbers in this section. **[MEASURED]** All values above are computed on a radial domain satisfying $R_{\text{max}} \geq 3n_{\text{max}}^2$, the convergence rule established in Appendix A, and were obtained by two independent routes — exact grid-free evaluation of the Slater integrals in closed form, and converged quadrature on a graded mesh — agreeing to four decimal places on every leg. An earlier version of this section reported $K^{0.84}$, $\|T'\|_1^{\text{off}} \sim K^{1.05}$ and an exponent “trending toward 1”. Those were computed on a fixed 60-bohr radial domain, which truncates configurations whose orbitals extend to 100–200 bohr; because the truncation error grows with K (by a factor $1.011 \rightarrow 1.205$ across the fitted range) it inflated the fitted slope and reversed its apparent drift. The corrected values are reported here.

V. THE QUANTUM ALGORITHM AND ITS NOVELTY

Equations (3)–(6) describe an unusually clean target for quantum eigenvalue estimation: block-encode the fixed, bounded, pure-number matrix M ; estimate its eigenvalues p_κ by qubitization or phase estimation [9];

return the energies $E = -\frac{1}{2}p_\kappa^2$. **[OBSERVATION]** A targeted prior-art search finds no existing quantum algorithm combining (i) a Sturmian/hyperspherical basis, (ii) a quantum eigenvalue routine, and (iii) the isoenergetic inversion (fix E , solve for a scale). Coulomb–Sturmian bases are classical in the literature [5]; quantum electronic structure uses Gaussian and plane-wave bases; and quantum generalized-eigenvalue solvers [10, 11] run the forward direction. The closest relative [11] scans a candidate eigenvalue as a singularity test, which is not the isoenergetic advantage. The atomic algorithm therefore appears novel.

The two documented cost risks for such a scheme are, moreover, *absent in the atomic case*. The metric-conditioning multiplier of block-encoded generalized eigenproblems [10, 12] does not apply, because Eq. (3) carries no metric. The outer energy search that would otherwise multiply the cost does not arise, because the eigenvalues are the energies. What remains is a single figure of merit, the 1-norm of M , which Eq. (6) measures to grow more slowly than the matrix dimension over every computed basis (with the caveats on window and mechanism recorded there). We stress that this is a resource analysis, not a hardware demonstration.

VI. THE MOLECULAR FRONTIER: SHIBUYA–WULFMAN CONDITIONING

[INTERNAL THEOREM] *The general- V_0 form, and what the metric is.* Before specializing, the general statement costs two lines and is worth having, because it says what the molecular metric actually is. Projecting $\langle \Phi_\mu | H - E | \Phi_\nu \rangle = 0$ and substituting the Sturmian equation on the ket, $(-\frac{1}{2} \sum_j \nabla_j^2 - E) | \Phi_\nu \rangle = -\beta_\nu V_0 | \Phi_\nu \rangle$, gives

$$\sum_\nu \left(\langle \Phi_\mu | V | \Phi_\nu \rangle - \beta_\nu \langle \Phi_\mu | V_0 | \Phi_\nu \rangle \right) C_\nu = 0, \quad \text{that is} \quad \mathbf{V}_0 \mathbf{C} \mathbf{L}^2 \mathbf{V}_0 \mathbf{C} = \mathbf{V}_0 \mathbf{B} \mathbf{C}, \quad (11)$$

$\mathbf{B} = \text{diag}(\beta_\nu)$. Every L^2 overlap cancels *identically* — for any local V_0 , and whether or not the configurations are orthonormal. **[MEASURED]** Atomically $\langle \Phi_\mu | V_0 | \Phi_\nu \rangle = -Z R_\nu \delta_{\mu\nu}$ and $\beta_\nu Z = p_\kappa / R_\nu$, so $\mathbf{V}_0 \mathbf{B} = -p_\kappa \mathbf{I}$ and Eq. (11) collapses to Eq. (3): verified to 6×10^{-10} , with the reconstruction $Z \text{diag } R_\nu - G$ bit-identical to the assembled M . **[OBSERVATION]** Molecularly $\mathbf{V}_0$ is *not* diagonal, so the matrix on the right-hand side is the weighting potential itself — not the identity, and not the L^2 overlap. That identifies the Shibuya–Wulfman matrix rather than merely naming it: for $V_0 = -\sum_A Z_A / |\mathbf{r} - \mathbf{R}_A|$ its off-diagonal entries are exactly the cross-center nuclear-attraction integrals, which is what `geovac/shibuya_wulfman.py` computes. The molecular metric is not an extra structure the method acquires under generalization; it *is* V_0 . *Provenance:* the Shibuya–Wulfman integrals and their momentum-space evaluation are Avery’s; the identification of the resulting

matrix *as* the V_0 -weighted overlap is our formalization — a targeted consultation could not locate that sentence in the texts, only the integrals it would describe.

[ESTABLISHED, from Avery] Molecularly there is no closed-form counterpart to $\beta_\nu Z = p_\kappa / R_\nu$. Because V_0 is multi-center, the weighting factors are obtained as the exact eigenvalues of the one-electron molecular Sturmian problem $[-\frac{1}{2}\nabla^2 - E + \beta_\nu V_0]\phi = 0$ at fixed E , with a many-electron configuration’s β_ν following from the isoenergetic condition on that spectrum. This is what makes the molecular orbitals of Sec. VI S_{SW} -orthonormal *by construction* rather than by arrangement: potential-weighted orthonormality is automatic for eigenvectors of a common Sturmian problem with distinct β , and S_{SW} is the potential doing the weighting. Two facts the paper had been carrying separately are one fact.

The clean atomic result does not fully transfer to molecules. The multi-center one-electron isoenergetic problem is again *generalized*,

$$\sum_\mu [W_{\mu'\mu} - k S_{\mu'\mu}] C_\mu = 0, \quad (12)$$

with S the Shibuya–Wulfman matrix [3], whose condition number multiplies the block-encoding cost: inverting or square-rooting S by quantum singular-value transformation costs a polynomial degree growing with κ (Sec. VIA) [13, 14]. In Hermitian form $S_{\mu'\mu} = (2k^2)^{-1} \langle \nabla \varphi_{\mu'} | \nabla \varphi_\mu \rangle + \frac{1}{2} \langle \varphi_{\mu'} | \varphi_\mu \rangle$.

[MEASURED] On a two-center s -orbital Coulomb–Sturmian basis we find: (i) the intra-center block of S is *exactly* the identity—in the momentum-space evaluation, diagonal $\{1, 1, 1\}$ with off-diagonal $\sim 10^{-16}$ (the 2D-grid values $\{1, 0.999, 0.998\}$ are discretization error)—where the L^2 overlap’s intra-center block is ill-conditioned (cond 5.8, itself growing with basis), so the within-center disaster of Sec. II is genuinely absent from S ; (ii) across two centers S is uniformly better-conditioned than the L^2 overlap, and markedly better at larger separation:

R (bohr)	basis	cond(L^2)	cond(S)
1.4	6	39.1	30.1
2.0	6	20.9	15.4
4.0	6	11.1	4.4

but (iii) $\text{cond}(S)$ still grows with basis size—*polynomially*, $\text{cond}(S) \sim N^{1.85}$, essentially the same rate as the L^2 overlap’s $N^{1.70}$ (decisively power-law, not exponential), with the largest eigenvalue bounded, $\lambda_{\max} < 2$, and all the growth in $\lambda_{\min} \rightarrow 0$ as the basis approaches linear dependence—driven by the inter-center coupling (off-block magnitude ~ 0.4 – 0.6), so the improvement over L^2 is only modest at bonding distance. All three cross-check across a 2D position-space grid, the closed-form momentum-space quadrature, and a direct three-dimensional momentum integral (agreement to $\sim 10^{-10}$).

[SYMBOLIC + MEASURED] The conditioning law is in fact *derived*, and the fitted exponent is a window reading. Because the intra-center block is exactly

the identity, $S = \begin{bmatrix} I & C \\ C^\dagger & I \end{bmatrix}$ has spectrum $\{1 \pm \sigma_k\}$ with σ_k the singular values of the cross block (the cosines of the principal angles [17–19] between the two center subspaces), so $\text{cond}(S) = (1 + \sigma_{\max})/(1 - \sigma_{\max})$ *exactly*. In the sine basis the cross block is the finite section of a multiplication operator with symbol $j_0(kR \cot(\chi/2))$, whose supremum 1 is attained at $\chi = \pi$; the band-limited concentration rate at a quadratic symbol maximum gives

$$1 - \sigma_{\max} = \frac{(kR)^2}{24} \frac{\pi^2}{n^2} + o(n^{-2}), \quad (13)$$

(n orbitals per center), i.e. the conditioning exponent is exactly 2 asymptotically, with the single-parameter collapse $(1 - \sigma_{\max})(n/kR)^2 \rightarrow \pi^2/24$ (confirmed to $\sim 1\%$ at $n = 160$; the leading-order derivation is asymptotic, the remainder is measured, not bounded). The fitted $N^{1.85}$ above—and the $N^{1.81}$ of the same quantity at $R = 2$ in the sprint record—are pre-asymptotic slopes of this one law read on different windows. The same singular spectrum carries the composition wall of the balanced-basis arc: the center-projection commutator is $\| [P_A, P_B] \| = \max_k \sigma_k \sqrt{1 - \sigma_k^2}$ [25, 26], which *saturates* at its algebraic ceiling $\frac{1}{2}$ (some σ_k lands near $1/\sqrt{2}$) as the spectrum densifies—the previously reported 0.50 is the saturation value—so the two walls are functionals of one object, with all of the severity information in $1 - \sigma_{\max}$ (`tests/test_paper60_sigma_law.py`).

The molecular metric cost is therefore *mitigated*, not *dissolved*: Eq. (12) retains a polynomial conditioning multiplier that is worst where the chemistry lives, and it cannot simply be transformed away—conditioning is orthogonal-invariant, so the momentum-space form (which cancels the Fock Jacobian to a clean one-dimensional quadrature) yields the *identical* matrix and is an exact, grid-free *evaluation* win, not a conditioning improvement. Two genuine conditioning levers do exist: at larger separation $\text{cond}(S) \rightarrow 1$ rapidly while the L^2 overlap cannot follow (its intra-center conditioning grows with basis regardless); and grade/ungrade symmetry adaptation exactly block-diagonalizes S , isolating a perfectly-conditioned grade sector ($\text{cond} \approx 2$, flat in basis and R) and confining all ill-conditioning to the ungrade sector—a free $\sim 2\times$ effective-cost reduction in the homonuclear case. The grade flatness is itself derived: for equivalent centers the blocks are exactly $I \pm C$, and the near-dependent direction has symbol value $+1$, so it lives entirely in $I - C$ (ungrade) while $\text{cond}(I + C) \rightarrow 2/(1 + \min_x j_0(x)) = 2.555041\dots$, an exact constant of the sinc symbol independent of R and of basis size (`tests/test_paper60_sigma_law.py`); the “ $\kappa \approx 2$ ” rows below are this constant read at small n .

[MEASURED] This lever is a property of *equivalent* centers, not of symmetry as such, and a probe on water (H_2O , C_{2v} : O + 2H) makes the distinction sharp. For an s -only shared-scale basis the only symmetry action is the H $\leftrightarrow$ H swap (O sits on the C_2 axis, fixed), so the totally-symmetric A_1 block—which holds the ground state—contains *both* the O functions and the symmet-

ric H combinations, and with them the O $\leftrightarrow$ H coupling between symmetry-*inequivalent* centers that the group cannot remove. Building water’s three-center SW metric exactly (every s - s block depends only on inter-center distance) and adapting to C_{2v} , the A_1 ground-state block’s conditioning *grows*, $\text{cond}(A_1) \sim N^{1.97}$ ($R^2 = 1.000$), essentially the raw metric ($19.9 \rightarrow 698$ over $N = 6 \rightarrow 36$)—*not* flat like the H_2^+ gerade sector ($\text{cond} \approx 2$ throughout). Zeroing only the O $\leftrightarrow$ H block collapses it back to $\text{cond} \approx 2$, pinning that inequivalent-center coupling as the sole driver. The lever therefore does not transfer to a polyatomic with a symmetry-unique heavy center; it is confined to homonuclear-diatomic (equivalent-center) systems and their close analogs. The water block obeys the same derived law: the O $\leftrightarrow$ (H + H) canonical correlations follow Eq. (13) with symbol curvature $c_{\text{sym}} = R_{\text{OH}}^2/24 - R_{\text{HH}}^2/96$ (exponent again exactly 2; the fitted 1.97 is the same pre-asymptotic reading), and $\text{cond}(A_1) \approx 1.12 \times (1 + \sigma_{\max})/(1 - \sigma_{\max})$ with a constant prefactor—the canonical correlation is the whole story. The structural reason the symmetry cannot help is now exact: O and H are not exchanged by any group element, so both $1 \pm \sigma$ eigenvector families remain inside A_1 and no block absorbs the divergent factor. A low-rank escape is also closed *structurally*: the coupling’s singular spectrum discretizes the range of the symbol (its effective rank grows linearly with basis size, participation ratio ≈ 7 at $N = 36$), and since $1 - \sigma_k \approx c_{\text{sym}}(k\pi)^2/n^2$, an exact rank- r (Woodbury) treatment of the top correlations divides the residual κ by $(r+1)^2$ only—a useful constant factor ($\approx 40\text{--}75\times$ in d_{inv} terms at $N = 12\text{--}36$) but never a change of exponent; flattening would need $r \propto n$, i.e. classically diagonalizing the whole coupling, the sparsity-destroying orthogonalization already excluded.

A. Resource estimate: the gerade lever is the equivalent-center payoff

We convert the measured conditioning into a block-encoding resource count for the one-electron H_2^+ secular equation Eq. (12), at chemical accuracy ($\epsilon_E = 1.6$ mHa) and $R = 2$ bohr. We whiten the generalized problem to a *standard* eigenproblem $\tilde{M} = S^{-1/2} W S^{-1/2}$ whose eigenvalues are the scales k ; the metric enters only through $S^{-1/2}$, applied by quantum singular-value transformation with a polynomial approximating $x^{-1/2}$ on $[\kappa^{-1}, 1]$ of degree $d_{\text{inv}} \sim \kappa \ln(\kappa/\epsilon)$ [13, 14], $\kappa = \text{cond}(S)$. This d_{inv} is the metric penalty—the quantitative content of the statement that the conditioning multiplies the block-encoding cost [13, 14]. Eigenvalue estimation to precision $\epsilon_k = \epsilon_E/k$ costs $Q \sim (\pi/2)\lambda_{\text{eff}}/\epsilon_k$ qubitization queries [9], with subnormalization $\lambda_{\text{eff}} = k_{\text{max}}$; each query calls one block-encoding of W and d_{inv} of S , so the metric multiplies the query count by $(1 + d_{\text{inv}})$.

[MEASURED] The algorithm binds: solving Eq. (12) for H_2^+ by isoenergetic self-consistency (build the basis at scale k , solve $[W - k'S]C = 0$, locate $k' = k$) gives

TABLE I. Block-encoding resource for the one-electron H_2^+ isoenergetic secular equation Eq. (12) at chemical accuracy ($\epsilon_E = 1.6$ mHa, $R = 2$ bohr), $N = 16$ ($n_{\max} = 8$ per center). d_{inv} is the QSVT degree for $S^{-1/2}$ (the metric penalty); qubits = $\lceil \log_2 N_{\text{dim}} \rceil + n_{\text{phase}} + n_{\text{anc}}$ with $n_{\text{phase}} = 10$. The qubitization query count $Q \approx 2.5 \times 10^3$ is common to every row (it depends on λ_{eff} and ϵ_k , not the metric). A resource model on the measured κ ; the $O(1)$ constant in d_{inv} cancels in the cross-metric ratios but not in the absolute counts.

metric	κ	d_{inv}	qubits
atomic (no metric, Sec. III)	1	0	23
SW gerade (σ_g ground)	2.3	18	26
SW full	91.8	1050	31
Gaussian LCAO (ratio 2)	1.99×10^4	3.3×10^5	31

the σ_g ground root $k = 1.475$, i.e. $E_{\text{elec}} = -1.088$ Ha against the reference -1.103 Ha—a 1.3% s -only under-binding (no p -polarization, the expected basis limitation, above the exact value)—and pins $\lambda_{\text{eff}} = k_{\max} \approx 1.7$.

[RESOURCE MODEL, on the measured κ] Table I gives the counts. Three readings: (i) the metric-free atomic case ($\kappa = 1$, $d_{\text{inv}} = 0$) is recovered as the zero-penalty limit; (ii) the full SW metric costs $d_{\text{inv}} \sim 10^3$ by $N = 16$; but (iii) the H_2^+ ground state σ_g is *gerade*, and the gerade sector’s conditioning is flat ($\kappa_g \approx 2$ across $N = 4$ –20), so the ground-state $S^{-1/2}$ costs $d_{\text{inv}} \approx 16$ –18 *independent of basis size*—a $\sim 60\times$ reduction versus the full metric at $N = 16$, on a register halved to $\lceil \log_2(N/2) \rceil$ qubits. The gerade lever is thus not merely the $2\times$ symmetry factor of the previous paragraph: it delivers the chemically relevant state at a metric penalty that *does not grow with basis size*. This is the equivalent-center payoff (the ground-state lever for homonuclear diatomics and their close analogs, not for a general polyatomic).

[RESOURCE MODEL] Against the same H_2^+ problem in a standard even-tempered s -Gaussian LCAO metric (ratio 2), the SW inverse-degree is 10^2 – 3×10^2 times smaller (full metric) and 10^3 – 10^4 smaller (gerade). This is a metric-versus-metric comparison and carries an honest caveat: the Gaussian conditioning is ratio-dependent—a denser ratio-1.6 set (approaching completeness) blows up as $\text{cond} \sim N^6$ ($\kappa \sim 10^5$ at $N = 16$), a sparser ratio-3 set grows only as $N^{1.4}$ but leaves coverage gaps and still carries $\sim 10\times$ the SW prefactor. This is the classic Gaussian coverage-versus-linear-dependence trade-off; the Coulomb–Sturmians evade it because they are a *complete* set at one scale whose metric is polynomially conditioned ($\text{cond}(S) \sim N^{1.85}$ on this window; derived asymptote N^2 , Eq. (13)) with a small prefactor. Production Gaussian simulation of course absorbs the overlap classically in a prior mean-field step; the metric penalty priced here is precisely the cost of the on-device angular generation that motivated the Sturmian encoding in the first place.

[MEASURED] The large- R lever is quantified alongside: $\text{cond}(S)$ falls to 2.2 at $R = 10$ bohr (at the sweep’s

$N = 12$, d_{inv} : $862 \rightarrow 17$ over $R = 1.4 \rightarrow 10$; at Table I’s $N = 16$ the $R = 1.4$ figure is larger still, so the lever is stronger than these numbers suggest), while the Gaussian metric stays $\sim 2 \times 10^3$ —its intra-center linear dependence does not relax with separation.

[MEASURED] Two probes price the derived law’s exploitability (resource-model tier; CHANGELOG v4.105.0). *Spectrum-aware inversion*—replacing the worst-case interval approximation for $S^{-1/2}$ by constrained interpolation on the *known* spectrum $\{1 \pm \sigma_k\}$ (classically computable by an $O(N^3)$ values-only SVD that touches no integral sparsity; the asymptotic law alone is 11–21% off at $\lambda_{\min}$ and does not suffice)—buys a constant factor ≈ 8 – $10\times$ at the *same* κ exponent, and the Bernstein $\Theta(\kappa)$ floor for *both* problems bounds it there (an argument, not a measurement: the floors hold for the interpolation and interval families alike, because the small eigenvalues are spaced at exactly the scale on which $x^{-1/2}$ varies). At this table’s $N = 16$ point the honest chain is model $1050 \rightarrow$ same-convention minimax $209 \rightarrow$ aware 27. An *exponent-changing* variant exists—a PSD shift of $[\lambda_{\min}, \lambda_{\max}]$ onto $[-1, 1]$, needing only Eq. (13)’s closed-form endpoints, gives degree $\approx 2.8\sqrt{\kappa}$ —but is gated by its LCU subnormalization (the naive 1-norm ≈ 3 squeezes the spectrum back to linear-in- κ , *worse* than baseline; with $\sim 3\times$ amplification the net at $\kappa = 864$ is $\approx 5 \times 10^2$ effective queries vs $\approx 1.9 \times 10^3$)—scoped, not priced beyond this. Separately, an *analytic momentum-space factorization* of the ERI tensor (the closed-form momentum densities as a continuous factorization index) matches but does not beat plain density fitting: eigen-Cholesky DF on the same tensor reaches within 13% of its 1-norm at rank $\leq n_{\text{orb}}^2$, versus 2.5 – 4.3×10^4 momentum leaves—the observed gains are DF’s, not the momentum representation’s.

B. Many-electron molecules: the metric does not compound

For a many-electron molecule the generalized-Sturmian secular equation is still $[T^0 + T' - p_{\kappa}\mathbb{I}]B = 0$ —a *standard* eigenproblem, the $-p_{\kappa}\mathbb{I}$ being the identity, not a metric (Avery’s general form [3]); only the atomic conveniences “ T^0 diagonal” and “ T' pure numbers” are lost. The resource-relevant question is whether the Shibuya–Wulfman metric penalty of Sec. VI *compounds* with electron number. It need not.

[SYMBOLIC + MEASURED] A k -electron configuration is a Slater determinant of one-electron molecular orbitals, so its configuration-space overlap is the k -th compound matrix [18, 20–22] of the one-electron overlap, with condition number $(\lambda_1 \cdots \lambda_k)/(\lambda_{N-k+1} \cdots \lambda_N)$ in the one-electron overlap spectrum $\{\lambda_i\}$. Built from the *raw* L^2 -normalized many-center Coulomb–Sturmians this *grows* with electron number—on the H_2 -geometry two-center basis ($N = 6$), $1\times, 8\times, 9\times$ $\text{cond}(S_{L_2})$ at $k = 1, 2, 3$ (worst at half-filling), bounded

by $\text{cond}(S_{L^2})^{\min(k, N-k)}$ and driven by the near-linear-dependence eigenvalue. Built instead from the *molecular Sturmians*—the Shibuya–Wulfman eigenvectors, S_{SW} -orthonormal by construction—the configuration overlap is the *identity at every electron number*—an algebraic consequence of the compound-matrix identity, not a numerical finding; only the $1 \times, 8 \times, 9 \times L^2$ figures above are measured. The metric is therefore paid once, at the one-electron molecular-orbital construction ($\text{cond}(S_{\text{SW}}$) above, with the gerade lever), and the N -electron secular equation inherits an identity metric: **the molecular metric penalty is one-electron, not N -electron**. It is the on-device analogue of the once-executed classical mean-field step that lets production Gaussian simulation carry no metric into the many-body Hamiltonian.

[MEASURED] The compound-matrix identity also fixes *where* the two centers decompactify: because a k -electron configuration overlap is the k -th compound of the one-electron overlap, the many-electron decompactification front moves only through screening (the occupied decay length), while correlation moves a distinct signed cross-center coherence front through the antibonding occupation. Both fronts, with their measured H_2 and HeH^+ tests, are developed in Paper 58 [15] (Sec. “the continuous side,” the many-electron front); they are bonding physics, not encoding cost, and ride in here only on the shared compound-matrix identity.

[MEASURED] The interelectron matrix T' is assembled from two-center electron-repulsion integrals, and these are in hand. We evaluate the shared-scale Coulomb–Sturmian two-center ERIs ($pq|rs$) numerically (a single multipole route about one nucleus, so every center class is uniform) and validate them against the framework’s *exact* closed form: the one-center ($1s|1s|1s|1s$) = $\frac{5}{8}$ to 4×10^{-6} , and the two-center ($AA|BB$) to $\sim 10^{-4}$ against `aabb_value` at $R \in \{1.4, 2.0, 3.0\}$ (the closed forms themselves are weight-one, π -free $\{E_1, \ln, \gamma\}$ [15, 16]). With these, the interacting method binds a molecule: minimal-basis ($1s$ on each center, common scale) H_2 in the metric-free molecular-Sturmian basis has an interior minimum at $R \approx 1.6$ bohr with $E_{\text{tot}} \approx -1.09$ Ha—the single- ζ minimal-basis value, underbinding the exact -1.174 Ha as s -only single- ζ must, above the exact energy throughout.

[OPEN, with the obstruction identified] What remains is the block-encoding 1-norm of the full N -electron secular matrix $T^0 + T'$, and here the molecular case parts from the atomic one at a structural level, not merely a computational one. The atomic sublinearity of Eq. (6) rides on the *clean diagonal* $T^0 = ZR_\nu$, which exists only because (i) the Goscinskian orbitals scale *with* p_κ (charge $Q_\nu = p_\kappa/R_\nu$), so the p_κ in $\langle 1/r \rangle$ cancels the $-p_\kappa^{-1}$ prefactor and M is a p_κ -independent linear matrix, and (ii) on a *single* center $\langle 1/r \rangle = Q/n^2$ collapses the collective scale to the root-sum-of-squares $T_{\text{bare}}^0 = \sqrt{\sum_p k_p^2} = ZR_\nu$ (we verify: for $\text{He } 1s^2$, $\sqrt{k^2 + k^2} = 2\sqrt{2} = ZR_\nu$ gives the exact bare -4 Ha; a naive sum of

scales gives -8). On *two* centers the nuclear attraction $\langle 1/r_A \rangle + \langle 1/r_B \rangle$ acquires off-center terms depending on $Q_\nu R$, so T^0 is neither diagonal nor p_κ -independent, and a fixed-scale molecular-Sturmian shortcut is not the isoenergetic matrix at all—it fails the non-interacting limit (returning $p_\kappa = 2k_{\sigma_g}$ where the correct value is $\sqrt{2}k_{\sigma_g}$). A faithful molecular 1-norm therefore requires the p_κ -scaled molecular Goscinskian (config charge $Q_\nu = p_\kappa/R_\nu$, two-center nuclear T^0 and two-center ERIs at the config scale, solved self-consistently; validated at $R \rightarrow \infty = 2$ H atoms and $R \rightarrow 0 = \text{He}$). The obstruction is thus not that the integrals are hard—they close (Papers [15, 16])—but that the single-center diagonal- T^0 structure that *produced* the sublinearity does not transfer.

[MEASURED] We can nonetheless price the practical alternative. Building the two-center mixed-charge integrals (overlap, nuclear attraction, and ERIs, validated against the exact closed forms and against $\langle 1s|1/r_B|1s \rangle$ to $\sim 10^{-4}$) and a full two-electron CI in the Loewdin-orthonormalized Sturmian basis yields a *validated* molecular solver— H_2 dissociates correctly to two H atoms ($E_{\text{tot}} \rightarrow -1.0$ Ha at $R = 6$) and binds above the exact -1.174 Ha at R_{eq} . Its *standard* block-encoding 1-norm $\lambda = \sum_{pq} |h_{pq}| + \sum_{pqrs} |(pq|rs)|$ grows polynomially, $\lambda \sim n_{\text{orb}}^{2.2}$, with no sublinear behavior—the atomic advantage of Eq. (6) is a property of the isoenergetic secular matrix, not of the basis, and it does not survive the loss of the diagonal T^0 . For molecules the resource question therefore answers in the ordinary way: the block-encoding cost scales polynomially, and the metric levers of Secs. VI A–VI B (the gerade sector, large- R , one-electron-only metric) are the real savings, not a sublinear matrix.

VII. CONCLUSION AND SCOPE

The message is a sharp split. *For atoms*, the isoenergetic generalized-Sturmian reformulation converts electronic structure into a standard, metric-free eigenproblem—a clean, apparently novel quantum secular equation in which the two feared cost risks of the genre are structurally absent. Its block-encoding 1-norm grows more slowly than the matrix dimension over every basis we can compute, and increasingly so. But the sharper statement is a trade-off rather than an efficiency: that growth belongs to the fixed- l_{max} basis-growth rule, reverses under full-shell growth, and is paid for by an accuracy floor whose mechanism is Eq. (10): the metric-free form exists only at the scale $\lambda = p_\kappa$, and that scale is not the variational optimum. The floor is a *ground-state* pathology — 6.8 mHa there at $K = 452$, but 1.72 mHa for 2^1S at identical encoding cost, and falling by 4.3 $\times$, 3.0 $\times$ and 2.6 $\times$ across the first three rungs of the 1S ladder. Read as a ground-state route to chemical accuracy it is not one; read as a cheap fixed-accuracy tier, and especially as an *excited-state* method — which is where the generalized-Sturmian

literature has always been strongest — it is a real operating point. The open question we leave is whether a weighting potential of different radial shape can lower the ground-state floor without breaking Eq. (9). *For molecules*, the Shibuya–Wulfman metric returns; it is genuinely better-conditioned than the L^2 overlap—its intra-center block is the identity—and much better at large separation, but its inter-center coupling still costs conditioning that grows with basis size. The one qualification that lifts this from a caveat to a payoff is symmetry: because the H_2^+ ground state is gerade and the gerade sector is perfectly conditioned and flat in basis size, the ground-state block-encoding pays a metric penalty that does not grow (Table I)—the growth is confined to the excited/ungerade states—though this is an *equivalent*-center property: a probe on water (C_{2v}) shows a symmetry-unique heavy center reinstates the growth in the ground-state (A_1) block (Sec. VI), confining the lever to homonuclear-diatomic-like systems. And the penalty does not compound with electron number: the Shibuya–Wulfman metric is discharged once, at the one-electron molecular-orbital construction, after which the many-electron secular equation carries an identity metric (Sec. VIB)—the metric cost is one-electron, not N -electron. The atomic case is a result; the molecular case is a frontier with a well-conditioned ground-state corner.

We have not run the algorithm on hardware, and we do not claim a resource advantage over production quantum chemistry: the systems where the encoding is demonstrably cheap are small. What we claim is structural and measured—that the generalized-Sturmian method’s classical elegance (no self-consistency, energies direct, pure-number matrices) is also, for atoms, a near-ideal quantum encoding—and that this reframes the on-device-generation intuition from a hope into a characterized boundary, clean on one side and open on the other.

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only its reading as a quantum-encoding target and the resource measurements. Developed with an AI-augmented workflow under principal-investigator direction, with the headline numerical claims validated against analytical benchmarks and a regression test suite.

Appendix A: Radial-domain convergence rule

[MEASURED] The Goscinskian configuration $\nu = (l, n_a, n_b)$ carries weighted charge $Q_\nu = p_\kappa/R_\nu$ with $R_\nu = \sqrt{n_a^{-2} + n_b^{-2}}$, and its orbital n is hydrogenic at exponent $a = Q_\nu/n$. Its radial density therefore peaks at $r_{\text{pk}} = n/a = n^2 R_\nu$, and the mismatched pairs $(l, 1, n)$ at $Q_\nu \approx 1$ reach furthest. A radial domain must satisfy

$$R_{\text{max}} \geq 3 \max_{\nu} \max_j n_j^2 R_\nu \approx 3 n_{\text{max}}^2 \quad (\text{A1})$$

for the 1-norm exponents to be converged. Measured relative error of $\|T'\|_1^{\text{off}}$ against a $R_{\text{max}} = 5n_{\text{max}}^2$ reference: a fixed 60-bohr domain gives 1.1×10^{-2} at $K = 74$ rising to 2.1×10^{-1} at $K = 340$; $c = 1$ gives $\sim 10^{-2}$ *falling* with K ; $c = 2$ gives $\sim 10^{-4}$; $c = 3$ gives $\sim 10^{-7}$. Exponents are converged to ± 0.001 at $c = 2$ and $\pm 10^{-4}$ at $c = 3$.

The distinction that matters is not the size of the error but its *sign of drift*. A domain that scales with the basis produces an error that shrinks with K and biases a log-log slope negligibly; a fixed domain produces an error that *grows* with K , i.e. with the fit variable, and biases the slope upward. The mechanism is normalization: an orbital truncated at 60 bohr retains as little as 2.7% of its norm at $n = 14$ and is then renormalized to unity, manufacturing a compact pseudo-orbital whose repulsion integral is $O(1)$ where the true coupling decays as n^{-2} . Equal- n configurations and $1s^2$ are unaffected, so single-configuration checks — including $E(1s^2) = -729/256$ and the $(5/8)Q$ unit-charge integral — are blind to it.

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